

- **Full Moon** This occurs on the 14th day after the New Moon. The Moon at this time shows its fully lighted surface. (Full Moon = Complete Moon)
- When two full Moons occurs in a single month, the second full Moon is called a **Blue Moon**.

Stars

- Stars are heavenly bodies made up of hot burning gases, thus shine by their own light. If the star is bigger than the Sun but not more than twice as big, it will turn into a **Neutron star** or **Pulsar**. They are formed due to novae and supernovae explosion.
- A star's colour indicates the temperature of its surface. Blue colour denotes maximum temperature. Then comes yellow, then red etc.

Evolutionary Stages of a Star

- Adult Star** It is the stage, where the helium core becomes increasingly heavy, accompanied with expanding outer layers.
- Red Giant** This stage results into swelling and reddening of the outer regions of the star. Such star of gigantic dimension is called Red star.
- White Dwarf** If the mass of the star is relatively small like that of our Sun, the gases that reach the outer layer are expelled. As these expelled gases cool and contract, the star becomes a white dwarf.

THE EARTH

- It is the 5th largest planet of our solar system. Only planet of our solar system where life exists. It is also called '**Blue Planet**'. It has circumference of 40008 km (polar) and equatorial circumference 40075 km and area of nearly 510 million sq kms.
- **Perigee** Nearest position of Earth to Moon (356000 km).
- **Apogee** Farthest position of Earth from Moon (407000 km).
- **Aphelion** Farthest position of Earth from Sun on 4th July, about 152 million km.
- **Perihelion** Nearest position of Earth to Sun on 3rd January, about 147 million km.
- The history of the Earth is studied in terms of geological years, periods and the epochs.

Shape of the Earth

The Earth is an oblate spheroid, a sphere flattened along the axis from pole to pole such that there is a buldge around the equator. Earth bulges at the equator and is flattened at the poles. The shape of Earth is also called **Geoid**. Geodesy is a branch of science which deals with measuring the Earth's size and shape.

The Earth's Movement

The Earth moves in space in two distinct ways rotation and revolution, which are as follow

- It rotates on its own axis from West to East once in every 24 hrs. It causes day and night.
- It revolves around the Sun in an orbit once in every 365¼ days. It causes the seasons and the year.

Rotation of the Earth

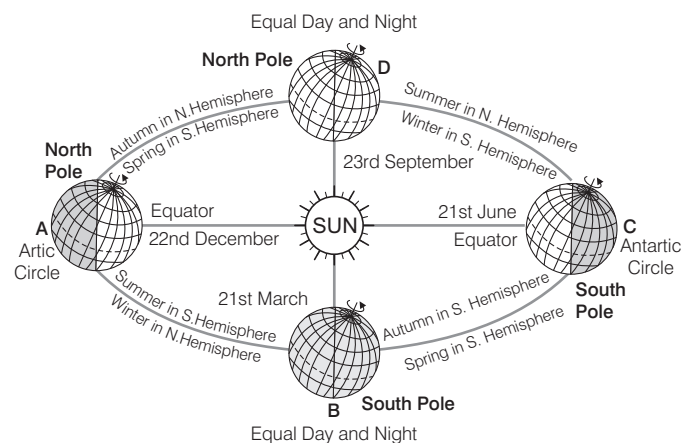
- Earth spins on its imaginary axis from West to East in 23 hrs, 56 min and 40.91 sec. The rotational speed at equator is maximum (1667 km/hr) and then decreases towards the poles, where it is zero.
- The rotation of the Earth causes day and night, change in the direction of winds and ocean currents, rise and fall of tides everyday and a difference of 1 hr between the two meridians which are 15° apart.
- The division between day and night over the Earth is considered by the circle of illumination. The circle of illumination always bisect the equator and divides Earth into light and dark halves i.e. a day hemisphere and a night hemisphere.

Revolution of the Earth

- It is the Earth's motion in elliptical orbit around the Sun. It takes 365 days, 5 hrs, 48 min and 45.51 sec. It leads to one extra day in every fourth year.
- The revolution of the Earth results in*

- change of season.
- variation of the length of the days and nights at different times of the year.
- shifting of the wind belts.

The figures below shows the revolution of the Earth and its effects on seasons and the variations of lengths of day and night.



Equinoxes

- These are the days, when days and nights are equal. Under this situation, the Sun is vertically overhead at the equator. It happens on 2 days of the year i.e. 21st March and 23rd September.
- 21st March **Vernal equinox** and 23rd September **Autumnal equinox**.

Summer Solstice

- After the March equinox, the Sun appears to move Northward and is vertically overhead at the Tropic of Cancer on 21st June. This is known as **Summer Solstice**.
- On 21st June, the Northern hemisphere will have its longest day and shortest night. The Southern hemisphere will have shortest day and longest night.

Winter Solstice

On 22nd December, the Sun is overhead at the Tropic of Capricorn and this is the winter solstice. Southern hemisphere will have its longest day and shortest night.

Latitude

- Latitude is the angular distance of a point on Earth surface from the centre of Earth, measured in degree. These lines are called **parallels of latitude** and on the globe they are circles. The circumference of the circles decreases from equator to pole and at the pole it converge to a point.
- The distance between any two parallels of latitude is always equal. One degree latitude = Approx 111 km
- The most important lines of latitudes are equator (0°), the Tropic of Cancer ($23\frac{1}{2}^\circ\text{N}$), the Tropic of Capricorn ($23\frac{1}{2}^\circ\text{S}$), the Arctic Circle ($66\frac{1}{2}^\circ\text{N}$) and the Antarctic Circle ($66\frac{1}{2}^\circ\text{S}$).

Longitude

- Longitude is the angular distance of a point on the Earth surface along the equator, East or West from the **Prime meridian**. On the globe, they form semi circles from pole to pole passing through the equator.
- Prime meridian is the semi circle from pole to pole, from which all the other meridians radiate Eastwards and Westwards up to 180° . In 1884, it was decided that the zero meridian is one that passes through the Royal Astronomical Observatory at **Greenwich** near **London**.
- 180° meridian (International Date Line) is exactly opposite to the Prime meridian. Such points are called Antipodal Points.

Universal Time (Standard Time) and Time Zones

- The Earth is divided in 24 longitudinal zones, each being 15° or 1 hrs apart in time ($360^\circ = 24$ hours, $360/24 = 15^\circ$ in 1 hrs or 1° in 4 minute) called Standard Time Zones.
- Larger countries such as USA, Russia and Canada, which have greater East-West stretch have to adopt several time zones for practical purposes.
- Russia has as many as 11 time zones.
- Both USA and Canada have five time zones, viz the Atlantic, Eastern, Central, Mountain and Pacific time zones.
- To avoid confusion about having many local times within one country, a particular meridian is chosen for the whole country, whose time is known as Standard Time.
- The Indian Government has accepted the meridian of 82.5° East for standard time, which is 5 hrs 30 min ahead of the Greenwich Mean Time GMT and is situated at Naini near Allahabad.

INTERNATIONAL DATE LINE

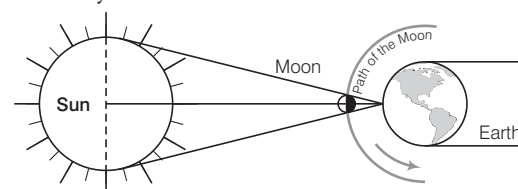
- It is the 180° meridian running over the Pacific ocean deviating at Fiji, Samoa and Gilbert Islands. This meridian is considered to be deviated at the land masses, so that the travellers do not feel inconvenient.
- One who is crossing the Date Line from West to East repeats a day and the one who travellers crossing it from East to West losses a day.

ECLIPSES

- An eclipse occurs when the Sun, Moon and Earth are in a straight line. There are two types of eclipses-Solar and Lunar eclipse.
- Generally, a total of seven eclipse including solar and lunar eclipses, take place in a year.

Solar Eclipse

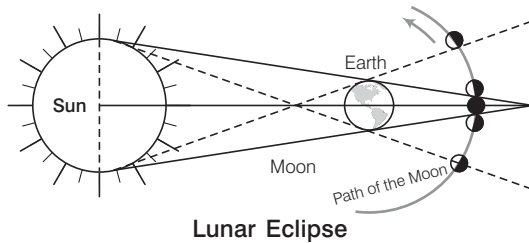
- Solar eclipse occurs when Moon comes between the Sun and the Earth. It can be partial or total.
- It happen only on a new Moon day when the Moon is in line with the Sun. A solar eclipse doesn't occur on every new Moon day.



Solar Eclipse

Lunar Eclipse

Lunar eclipse occurs when the Earth comes in between the Sun and the Moon. It occurs only on a full Moon day but not on every full Moon day because the Moon is not in the same position in relation to that of the Earth and the Sun on every full Moon day.

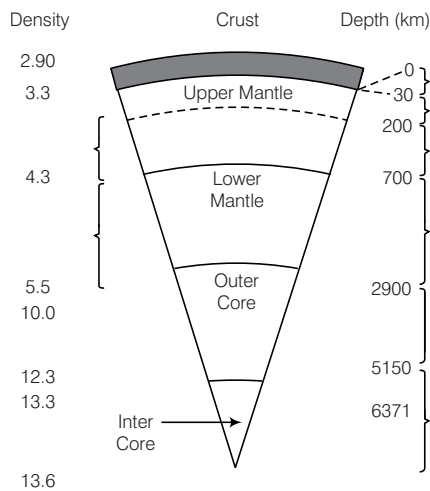


STRUCTURE OF EARTH'S INTERIOR

The interior of Earth is divided into three parts

The Crust

- The crust is the outermost and the thinnest layer of the Earth. This layer has the least density and its thickness varies about 8 to 40 km. **Mohorovic or Moho Discontinuity**, marks the lower limit of the crust and this discontinuity is identified on the basis of rock density.
- The rocks forming the crust of the Earth are rich in lighter minerals like silica and aluminium. Hence, this layer is also called as **Sial** (silica and aluminium). The average density of this layer is 2.7 gm/cm^3 .



The Mantle

- It is about 2900 km in thickness, composed of minerals in a semi solid state. It is further divided into two layer upper mantle and lower mantle. The upper part of the mantle is called **Asthenosphere**, which is about 250 km thick.
- Due to presence of minerals like silica and magnesium in the rocks forming this layer of the Earth, the mantle layer is also known as **Sima** (silica and magnesium). The average density of this layer is about 5.68 gm/cm^3 .
- The transitional zone separating the mantle from the core is called the **Gutenberg Discontinuity**.

The Core

- The core is the innermost layer of the Earth and occupies its center. It is about 3500 km in radius. This layer is also known as **Nife** (nickel and iron), because this layer contain large concentration of iron and nickel.
- Density of this part of the Earth is 17.2 gm/cm^3 and is many times greater than the average density of the Earth (5.53 gm/cm^3).

DISCONTINUITIES

The various layers are separated by discontinuities, which are evident in seismic data.

- Conrad** discontinuity lies between upper crust and lower crust.
- Mohorovicic** discontinuity between crust and mantle.
- Repetti** discontinuity lies between upper mantle and lower mantle.
- Gutenberg** discontinuity between core and mantle. Here, the Earth's density as well as velocity of P waves increases.
- Lehmann** discontinuity divides upper core and lower core.

Composition of Whole Earth

Iron (35%), Oxygen (30%), Silicon (15%), Magnesium (13%), Nickel (2.4%), Sulphur (1.9%).

Composition of Earth's Crust

Oxygen (46%), Silicon (28%), Aluminium (8%), Iron (6%), Magnesium (4%), Calcium (2.4%), Potassium (2.3%), Sodium (2.1%).

Continental Drift Theory

- This theory was given by **Alfred Wagener**, in 1915, to explain the origin and evolution of the continents and the oceans.
- According to this theory, about 250 million years ago, there was only one continent named **Pangea** means All Earth and it was surrounded by one mass of water body, named **Panthalassa**.
- The present shape of the continents and oceans is due to the break up of Pangea. This breaking process started about 200 million years ago.

- The Northern rift cuts Pangaea from East to West creating **Laurasia** in the North and **Gondwanaland** in South. A shallow sea called **Tethys** was situated between the Laurasia and the Gondwanaland.

Plate Tectonics Theory

- The theory of Plate tectonics states that the lithosphere is divided into several rigid segments, which include both oceanic and continental crusts. These segments are called **plates** and they are moving on the asthenosphere, which is not a liquid, but a solid which flows under stress.
- About 20 such plates have been identified. There are seven major plates Eurasia, Antarctica, North America, South America, Pacific, African and Indian Plate.
- Most of the Earth's seismic activity, volcanism and mountain building occur along with these dynamic boundaries.

Depending upon the type of movement, plate margins are of three types

- Divergent Plate Margin** (Constructive margins)
 - Convergent Plate Margin** (Destructive margins)
 - Parallel Plates Margin** (Conservative margin or Transform boundary)
- Collision can occur between two oceanic plates, one oceanic and one continental plate or two continental plates.

Due to collision three types of plate boundaries appears, which are as follow

Divergent Plate Boundary	Convergent Plate Boundary	Transform Fault Boundary
When the plates move apart with the upwelling of material from the mantle, divergent plate boundary results.	A convergent plate boundary is one, where two plates collide, one plate bending downward and subducting below the other.	They are located, where plates slide past one another without the creation or destruction of crust.
Formation of the mid oceanic ridges are the example of divergent plate margin.	Deep oceanic trench is formed adjacent to the zone of subduction. Fold mountains are the result of convergent plate boundary.	San Andreas Fault along the West coast of Mexico is a famous transform fault.
Example Surtsey Island was born along with mid-Atlantic ridge.	Example Himalayan boundary fault, Pacific ring of fire.	Example San Andreas Fault along the Western coast of USA.

Forces Affecting Earth's Crust

Study of forces affecting the crust of the Earth is very important as these forces and resultant movements are responsible for creation, modification and destruction of various landforms over the Earth's surface.

These forces are classified into endogenetic and exogenetic forces

Endogenetic Forces

The forces coming from within the Earth are called Endogenetic Forces. The heat inside the Earth is primary driving force of these forces. *These forces are further classified into two categories*

- Sudden or Catastrophic Forces** These forces act very swiftly and often in an unpredictable manner. Earthquake and volcanoes are major sudden forces.
- Slow or Diastrophic Forces** These forces act in an imperceptible manner where changes are discernible after millions of year. These forces are further classified into tensional and compressional forces.

Exogenetic Forces

These forces and processes occur over the Earth's surface and modify the landforms from outside. Erosion and weathering are two major phenomena of denudation of geomorphic features by exogenetic forces.

EARTHQUAKES

- It refers to the vibration of the Earth's surface caused by the endogenetic forces. The magnitude or intensity of energy released by an earthquake is measured by the **Richter Scale**, whereas the damage caused is measured by modified **Mercalli Intensity Scale**.
- The place of origin of earthquake is called **focus**. The place on the ground surface, which is perpendicular to the focus or hypocentre is called **epicentre**. **Seismology** is the special branch of geology that deals with the study of earthquake.
- *The waves generated by earthquake are called seismic waves and they are classified into three types :*
 - Primary Wave (P Waves)** These are the waves of short wavelength and high frequency. They are longitudinal waves and can travel through solid, liquid and gases.
 - Secondary Wave (S Waves)** These are the waves of short wavelength and high frequency. They are transverse waves, which travel through all solid particles.
 - Surface Waves or Long Waves (L Waves)** They are the waves of long wavelength, confined to the upper layer of the Earth's crust. It causes most of the earthquake's structural damage.
- Imaginary lines on map connecting the places where the seismic wave arrives generated by an earthquake, at the same time are known as **Homoseismal Lines**.

Distribution of Earthquakes

Most of the world earthquake occurs in

- The zones of young fold mountain, zones of folding and faulting, junction of continental and oceanic margin, zone of active volcanoes and along different plate boundaries.

The traditional zones of earthquake are

- Circum Pacific belt, Mid Continental belt and Mid Atlantic belt.

VOLCANISM

- Volcanism includes all phenomena connected with the movement of heated material from the interior towards the surface of the Earth.
- A volcano is a vent or opening, through which heated materials consisting of gases, water, liquid lava, fragments of rocks are ejected from the highly heated interior to the surface of the Earth.
- Volcanic eruptions are closely associated with several integrated processes such as gradual increase in temperature with increasing depth, due to the heat generated by degeneration of radioactive elements inside the Earth.
- Origin of magma due to the lowering of the melting point caused by reduction in pressure of overlying rocks due to fractures caused by splitting of plates.

Classification of Volcanoes

On the basis of mode of eruption :

Central Eruption or Explosive Type Here, the magma comes with great force through the small vent and spread to a distant position. e.g., Hawaiian type, Strombolian type, Pelean type, Vesuvius type, Volcanion type.

Fissure Eruption or Quiet Eruption Large quantities of lava quietly flow up from fissures and spread out over the surrounding areas. Successive flow of lava results in the growth of lava plateau. e.g., Deccan Plateau.

On the basis of periodicity of eruptions :

Active Volcano Volcano which erupt periodically. e.g., Etna, Stromboli, Mayon

Dormant Volcano Volcano which become quiet after their eruption for sometime. e.g., Fujiyama, Krakatoa, Barren Island

Extinct Volcano They have no indication of future eruption.

Various Volcanic Belts

Circum-Pacific Belt (Fire girdle of the Pacific or the fire ring of the Pacific). It extends across the Kamchatka Peninsula, Kurile Islands, the Islands of Japan, Philippines, New Guinea, New Zealand and the Solomon Islands.

Highest Volcanic Peaks Cotopaxi (South America), Fujiyama (Japan), valley of ten thousand smokes (Alaska).

Mid-Continental Belt (Volcanic zones of convergent continental plate Margins). It includes volcanoes of Alpine mountain chain, the Mediterranean sea and the fault zone of Eastern Africa of Stromboli, Vesuvius, Etna, Kilimanjaro etc.

Mid-Atlantic Belt The volcanoes are fissure eruption type e.g. Iceland, Canary Islands, Cape Verde, Azores etc.

ROCKS

The solid part of the Earth's crust are called rocks. The rocks are made up of two or more minerals.

Rocks are classified in three main types depending on the process of their formation

1. Igneous Rocks

- Formed due to the cooling, solidification and crystallisation of hot and molten magma. They are called as the **primary rocks** as all the other rocks are formed directly or indirectly from the **igneous rocks**.
- It is believed that the igneous rocks were formed during each period of geological history of the Earth. They are hard, granular and crystalline rock, less affected by chemical weathering. Moreover, it does not have any fossil or does not form any strata or layers of lava.

Classification of Igneous Rocks

On the basis of mode of occurrence :

- **Intrusive Rocks** They are formed due to the solidification of rising magma below the surface of the Earth. e.g. Granite, Lapolith, Batholiths, Sills etc.
- **Extrusive Rocks** They are formed due to cooling and solidification of hot and molten magma at the Earth surface. e.g. Basalt, Gabbro etc.
- *On the basis of silica content :*
 - **Acidic** It has more silica content e.g. **Granite**.
 - **Basic** It has less silica content e.g. **Gabbro**.

2. Sedimentary Rocks

- It is formed due to the aggregation and compaction of sediments derived from the older rocks, plants, animals and contains fossils of plants.
- Sedimentary rocks can be classified on the basis of the nature of sediments as mechanically, chemically and organically formed rocks.

3. Metamorphic Rocks

- These are the changed form of igneous and sedimentary rocks. These are the rocks, which change either in form or composition without disintegration.
- Already formed metamorphic rocks are metamorphosed and this process is called Metamorphosis. The agents of metamorphism are heat, compression and solution.